[Thm. Riemann's Theorem.
Let f be a complex function having an isolated singularity at $z=z_0$.
If f is bounded for some deleted neighborhood of z_0,
then z_0 is removable.
Prof. By the hypothesis, for some $R>0$, f is analytic and bounded in

$$0<|z-z_0|<R.$$

Given $z_1 \in B_R(z_0)\setminus\{z_0\}$, let $r>0$ s.t. $r<|z_1-z_0|<R$. So, f is analytic and bounded in the annulus between $C: |z-z_0|=R$ and $C_1: |z-z_0|=r$.
By the Cauchy's integral formular,

$$f(z_1)=\frac{1}{2\pi i}\cdot\left[\int_C \frac{f(w)}{w-z_1}\,dw-\int_{C_1}\frac{f(w)}{w-z_1}\,dw\right]$$

Meanwhile, by the ML inequality,

$$\left|\int_{C_1}\frac{f(w)}{w-z_1}\,dw\right| \le \sup|f|\cdot 2\pi r\cdot\frac{1}{|z_1-z_0|-r} \to 0 \quad \text{as } r\to 0.$$

So, $f(z_1)=\frac{1}{2\pi i}\int_C \frac{f(w)}{w-z_1}\,dw$.

Since z_1 was arbitrary, we can denote that

$$f(z)=\frac{1}{2\pi i}\cdot\int_C\frac{f(w)}{w-z}\,dw \quad \text{for all } 0<|z-z_0|<R,$$

So, define $f(z_0)=\frac{1}{2\pi i}\cdot\int_C\frac{f(w)}{w-z_0}\,dw$, we obtain the result.

[Thm. Suppose that f has an isolated singularity at z_0. TFAE:
1) For some $m \in \mathbb{N}$, $\lim_{z \to z_0} (z-z_0)^m \cdot f(z) = A \in \mathbb{C} \setminus \{0\}$. (That is, z_0 is pole of order m).
2). $f(z) \to \infty$ as $z \to z_0$
3). $f(z) = g(z)/(z-z_0)^m$ for some analytic $g(z)$ and $m \in \mathbb{N}$, with $g(z_0) \neq 0$.

Proof. Suppose that there exists $m \in \mathbb{N}$ s.t. $\lim_{z \to z_0} (z-z_0)^m f(z)$ exists with non-zero.
Given $\varepsilon = |A|/2$, there exists $\delta > 0$ such that

$$|z - z_0| < \delta \Rightarrow |(z-z_0)^m \cdot f(z) - A| < |A|$$

So, using the triangle inequality, for $0 < |z-z_0| < \delta$,

$$|A| - |z-z_0|^m \cdot |f(z)| < \frac{|A|}{2}$$

$$\Rightarrow \quad \frac{|A|}{2\delta^m} < \frac{|A|}{2|z-z_0|^m} < |f(z)|$$

Thus, taking $\delta \to 0$, then $f(z) \to \infty$. 1) $\Rightarrow$ 2).
Suppose that $f(z) \to \infty$ as $z \to z_0$. Put $g(z) = 1/f(z)$ in $B_\delta(z_0) \setminus \{z_0\}$ for some $\delta > 0$, which satisfies $f(z) \neq 0$. (If no such neibor exists, then f is identically zero.)
Now, since $f(z) \to \infty$ as $z \to z_0$, so $g(z) \to 0$ as $z \to z_0$, so z_0 is removable to g.
Then, the Taylor expansion

$$g(z) = a_0 + a_1(z-z_0)^1 + \cdots \quad (|z-z_0| < \delta).$$

is valid. Since $g(z) \neq 0$ in $B_\delta(z_0) \setminus \{z_0\}$, so not all the coefficients are zero.
Let m be the smallest integer s.t. $a_m \neq 0$. ($m \geq 1$ because $\lim_{z \to z_0} g(z) = a_0 = 0$).
So that $g(z) = a_m(z-z_0)^m + a_{m+1}(z-z_0)^{m+1} + \cdots$

Now, $$\frac{1}{(z-z_0)^m f(z)} = a_m + a_{m+1}(z-z_0) + \cdots \to a_m \text{ as } z \to z_0,$$

that is, $(z-z_0)^m f(z) \to a_m^{-1}$ as $z \to z_0$.] 2) $\Rightarrow$ 3).
3) $\Rightarrow$ 2) is directly, the analytic g is continuous at z_0.

Isolated singularity at ∞.

If f is analytic for all $R<|z|<\infty$, for some $R>0$,
then we say that f has an isolated singularity at ∞.

For this $R>0$, observe that

$f(z)$ is analytic at $|z|>R$ if and only if $f(\frac{1}{z})$ is analytic at $|z|<\frac{1}{R}$

$f(z)$ has an isolated singularity at ∞ iff $f(\frac{1}{z})$ has an isolated singularity at 0.

Thm. Let f be a non-constant entire function. Then,

f is polynomial <u>if and only if</u> $f(z)\to\infty$ as $z\to\infty$.

Proof. Suppose that f is a polynomial. WLOG, let $f(z)=a_0+a_1z+\cdots+a_nz^n$. $(a_n\neq 0)$.

Then, $f(\frac{1}{z})=\frac{a_n}{z^n}+\frac{a_{n-1}}{z^{n-1}}+\cdots+\frac{a_1}{z}+a_0$ so that $\lim_{z\to 0} z^n\cdot f(\frac{1}{z})=a_n\neq 0$,

So, $f(z)$ has a pole at ∞, so $\lim_{z\to\infty} f(z)=\infty$.

Conversely, suppose that $f(\frac{1}{z})$ has a pole of order m at 0. That is,

$$\lim_{z\to\infty}\frac{f(z)}{z^m}=A\in\mathbb{C}\setminus\{0\}.$$

That is, for some $R>0$, $|z|\geq R \Rightarrow \left|\frac{f(z)}{z^m}-A\right|<1$

$$\Rightarrow \left|\frac{f(z)}{z^m}\right|<1+|A|$$

$$\Rightarrow |f(z)|<(1+|A|)\cdot z^m$$

So f is a polynomial of degree at most m.

[Singularities of rational functions.]

Let p and q be polynomials over $\mathbb{C}$, $p(z) = a_0 + a_1 z + \cdots + a_n z^n$, $q(z) = b_0 + \cdots + b_m z^m$ ($a_n, b_m \neq 0$).
and $f(z) = p(z)/q(z)$. Then,

$f(z)$ has a removable singularity at ∞
iff $f(\frac{1}{z})$ has a removable singularity at 0
iff for some $\varepsilon > 0$ and $M > 0$, $0 < |\frac{1}{z}| < \varepsilon \Rightarrow |f(\frac{1}{z})| \leq M$.
iff " $|z| > \frac{1}{\varepsilon} \Rightarrow |f(z)| \leq M$.
iff $\deg p(z) \leq \deg q(z)$.

$f(z)$ has a pole of order k at ∞
iff $f(\frac{1}{z})$ has a pole of order k at 0.
iff $f\left(\frac{1}{z}\right) = \frac{p(\frac{1}{z})}{q(\frac{1}{z})} = \sum_{i=-k}^{\infty} c_i \cdot z^i$ for some $\{c_i\}_{i=-k}^{\infty}$
iff $p(\frac{1}{z}) = \sum_{i=0}^{n} a_i \cdot \frac{1}{z^i} = \left(\sum_{i=0}^{m} b_i \cdot \frac{1}{z^i}\right) \cdot \left(\sum_{i=-k}^{\infty} c_i z^i\right)$
iff $n = m + k$, that is, the order $k = \deg p - \deg q > 0$.

Long division.

Consider a function represented by

$$f(z)=\left(\sum_{n=0}^{\infty}a_n z^n\right)\Big/\left(\sum_{n=0}^{\infty}b_n z^n\right)=\frac{a_0+a_1z+a_2z^2+\cdots}{b_0+b_1z+b_2z^2+\cdots}\qquad (b_0\neq 0).$$

Since $b_0\neq 0$, this f is analytic at $z=0$, so f can be written as

$$f(z)=\frac{a_0+a_1z+\cdots}{b_0+b_1z+\cdots}=c_0+c_1z+\cdots$$

so that

$$\left(\sum_{n=0}^{\infty}a_nz^n\right)=\left(\sum_{n=0}^{\infty}b_nz^n\right)\cdot\left(\sum_{n=0}^{\infty}c_nz^n\right),$$ in some neighbor of the origin, thus the uniqueness theorem gives

$$a_n=\sum_{k=0}^{n}b_k\cdot c_{n-k},$$ the Cauchy Product.

Example.

Let $f(z)=\frac{\pi\cdot\cot\pi z}{z^4}$. Then,

$$f(z)=\frac{\pi}{z^4}\cdot\left(\frac{1-\frac{\pi^2}{2!}\cdot z^2+\frac{\pi^4}{4!}\cdot z^4+\cdots}{\pi z-\frac{\pi^3}{3!}\cdot z^3-\frac{\pi^5}{5!}\cdot z^5+\cdots}\right)$$

$$=\frac{1}{z^5}\cdot\left(\frac{1-\frac{\pi^2}{2}z^2+\frac{\pi^4}{4!}z^4-\cdots}{1-\frac{\pi^2}{3!}\cdot z^2+\frac{\pi^4}{5!}\cdot z^4-\cdots}\right)$$

$1=1\cdot c_0 \Rightarrow c_0=1$

$a_1=b_0c_1+b_1c_0=c_1+0=0. \Rightarrow c_1=0.$

$a_2=b_0c_2+b_1c_1+b_2c_0=c_2+0+-\frac{\pi^2}{6}\cdot 1=-\frac{\pi^2}{2}\Rightarrow c_2=-\frac{\pi^2}{3}.$

$a_3=b_0c_3+b_1c_2+b_2c_1+b_3c_0=c_3+0+-\frac{\pi^2}{6}\cdot 0+0=0 \Rightarrow c_3=0.$

$a_4=b_0c_4+b_1c_3+b_2c_2+b_3c_1+b_4c_0=c_4+c_2\cdot-\frac{\pi^2}{6}+\frac{\pi^4}{5!}=c_4+\pi^4\cdot(1/18+1/120)=\frac{\pi^4}{24}.$

$\Rightarrow c_4=\pi^4\left(\frac{1}{24}-\frac{1}{18}-\frac{1}{120}\right)=\pi^4\cdot\left(-\frac{1}{45}\right)=-\frac{1}{45}\pi^4.$

Consequently, the principal part of f is

$$\frac{1}{z^5}-\frac{\pi^2}{3}\cdot\frac{1}{z^3}-\frac{\pi^4}{45}\cdot\frac{1}{z}.$$